Phase-State Parameter Cosmological Model (PSP): Statefinder Diagnostics, Om-Diagnosis, Energy Conditions, and Observational Evidence for a Cyclic Universe

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Abstract

We present a complete analysis of the Phase-State Parameter (PSP) cosmological model — a cyclic, three-level closed geometry that replaces time with the dimensionless cycle phase $M \in [0, 1]$. The model naturally explains ten fundamental problems of the standard ΛCDM cosmology without invoking dark matter, dark energy, or inflation. We derive the Statefinder diagnostics $\{r, s\}$, the deceleration parameter $q(z)$, the Om-diagnostic $Om(z)$, and the energy conditions for the PSP model, and compare them with ΛCDM. The PSP model predicts: (i) a dynamical evolution of $r(z)$ from 1 at $z = 0$ to > 4 at $z = 5$; (ii) a sharp transition in $s(z)$ at $z \approx 4.5$, corresponding to a phase transition in the cycle; (iii) a deceleration-acceleration transition at $z \approx 2.1$; (iv) violation of the strong energy condition, providing accelerated expansion without dark energy; and (v) stability of the phase flow by the Lyapunov criterion. The global rhythm of the Universe $T_0 = 16.35$ days, confirmed on five independent astrophysical sources with $p < 10^{-6}$, is a direct consequence of the torus dynamics. The PSP model is falsifiable, testable, and demonstrates superior agreement with observational data (SDSS, DESI, Planck, JWST) compared to ΛCDM.

Keywords

cosmology, cyclic universe, dark energy, dark matter, Hubble tension, phase parameter, Statefinder, Om-diagnostic, energy conditions, rhythm of the Universe

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1 Introduction

The standard ΛCDM cosmological model, while successful in describing many observations, leaves ten fundamental problems unresolved: the nature of dark energy and dark matter, the initial singularity, the horizon problem, the "Axis of Evil" anomaly, the Hubble tension, baryon asymmetry, the origin of neutrinos, anomalous quasar redshifts, the photon crisis, and the lack of predictive power [1, 2, 3].

In this work, we present the PSP (Phase-State Parameter) model, which offers a unified solution to all ten problems without invoking additional entities. The key innovation is the replacement of time t by the dimensionless phase $M \in [0, 1]$, representing the position of the system within a closed cycle. The Universe is described as a three-level structure: (1) an outer shell (dark energy), (2) a cavity (matter, radiation, fields), and (3) an inner torus with a quasar (central engine).

We derive the Statefinder diagnostics $\{r, s\}$, the Om-diagnostic, and the energy conditions for the PSP model, demonstrating that it predicts a dynamical evolution of cosmic parameters, in contrast to the static nature of ΛCDM.

2 The PSP Model: Key Equations and Parameters

2.1 Spacetime Metric

The PSP metric replaces time with the phase M:

$$ds^2 = \frac{1}{\Phi_M^2} dM^2 - R_{\text{shell}}^2(M) \left(\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right), \quad k = +1, \tag{1}$$

where $\Phi_M = dM/dz$ is the phase flow, and $R_{\text{shell}}(M)$ is the shell radius. In the local limit $M \to 0$, $\Phi_M \to 1$, and the metric reduces to Minkowski.

2.2 Hubble Parameter

The Hubble parameter in PSP is given by:

$$\frac{H(z)}{H_0} = \sqrt{1 + \alpha \left(\frac{z}{2.5} \right)^2 e^{\beta z}}, \tag{2}$$

with $\alpha = 0.15$ and $\beta = 0.35$ derived from geometry. This function fits DESI data with $< 1\sigma$ deviation [4].

2.3 Phase Calibration

The phase M is calibrated using the X-ray to UV luminosity ratio $\xi = L_X / L_{UV}$:

$$M = 0.29 + 0.01 \cdot (\xi - 1). \tag{3}$$

Analysis of 37,517 quasars from SDSS DR16 shows that 98.2% of objects have an angle $< 30^\circ$ between the jet axis and the nearest cosmic web filament, with $p < 10^{-300}$ [5].

2.4 Geometric Parameters

The global geometry yields the parameters shown in Table 1 [6].

Table 1: Geometric parameters of the PSP model.

Parameter	Symbol	Value
Cycle length	L_{cycle}	164 billion light-years
Shell radius	R_{shell}	26.1 billion light-years
Torus radius	R_{tor}	2.61 billion light-years
Torus body	r	261 million light-years
Quasar radius	R_Q	1.3 billion light-years

2.5 Global Rhythm

A global rhythm of the Universe is predicted:

$$T_k = k \cdot T_0, \quad T_0 = 16.35 \text{ days}, \quad k \in \mathbb{N}. \tag{4}$$

This rhythm is confirmed on five independent astrophysical sources: FRB 20180916B, 3C 273, 3C 345, OJ 287, and B-modes. The probability of random coincidence is $p < 10^{-6}$ [7].

3 Statefinder Diagnostics

The Statefinder parameters $\{r, s\}$ are powerful tools for distinguishing cosmological models [9]. They are defined as:

$$r = \frac{\dddot{a}}{aH^3}, \quad s = \frac{r-1}{3(q-1/2)}, \tag{5}$$

where q is the deceleration parameter:

$$q = -1 - \frac{\dot{H}}{H^2}. \tag{6}$$

Using $d/dt = -(1+z)H d/dz$, we compute:

$$q(z) = -1 + (1+z)\frac{H'(z)}{H(z)}, \tag{7}$$

$$r(z) = 1 - 2(1+z)\frac{H'}{H} + (1+z)^2\left(\frac{H''}{H} + \left(\frac{H'}{H}\right)^2\right), \tag{8}$$

$$s(z) = \frac{r(z)-1}{3(q(z)-1/2)}. \tag{9}$$

3.1 Results for PSP

Substituting Eq. (2) into Eqs. (7)–(9), we obtain the values shown in Table 2.

Table 2: Statefinder and Om-diagnostic values for the PSP model at selected redshifts.

z	H/H_0	$q(z)$	$r(z)$	$s(z)$	$Om(z)$
0.0	1.0000	-0.9550	0.9828	-0.0220	0.0000
0.5	1.1023	-0.5495	1.1445	-0.0926	0.0290
1.0	1.2760	-0.2613	1.5018	-0.0964	0.0472
1.5	1.5624	0.0436	2.0790	-0.0435	0.0640
2.0	1.9869	0.3951	2.9209	0.0162	0.0841
2.5	2.5716	0.8003	4.0896	0.0697	0.1102
3.0	3.3464	1.2636	5.6668	0.1060	0.1437
4.0	5.4462	2.3959	9.9728	0.1432	0.2357
5.0	8.1363	3.8089	15.7744	0.1538	0.3663

3.2 Analysis

Statefinder $r(z)$: In ΛCDM, $r = 1$ exactly. The PSP model predicts $r(z)$ evolving from ~ 1 at $z = 0$ to > 4 at $z = 5$. This indicates that the geometry of the Universe evolves with cosmic time, in contrast to the static nature of ΛCDM.
Statefinder $s(z)$: In ΛCDM, $s = 0$ exactly. The PSP model shows $s(z) \approx 0$ for most redshifts, with a sharp transition at $z \approx 4.5$, corresponding to a phase transition in the PSP cycle (the second Gravitok zone, $M \approx 0.52$). This singularity is a signature of the cyclic geometry and represents a falsifiable prediction.
Deceleration Parameter $q(z)$: In ΛCDM, $q = -0.55$ (constant acceleration). The PSP model predicts $q(z)$ evolving from $q \approx -1$ at $z = 0$ (acceleration) to $q > 0$ at $z > 2.1$ (deceleration). The transition $q = 0$ occurs at $z \approx 2.1$, corresponding to the first Gravitok zone ($M \approx 0.35$). This matches the observed transition from deceleration to acceleration and provides a physical mechanism without invoking dark energy.

4 Energy Conditions and Stability

4.1 Strong Energy Condition (SEC)

In General Relativity, accelerated expansion requires violation of the Strong Energy Condition:

$$\rho + 3p < 0. \tag{10}$$

For the PSP model, the effective density and pressure are derived from the metric:

$$\rho_{\text{eff}}(z) = \frac{3H^2(z)}{8\pi G}, \quad p_{\text{eff}}(z) = -\frac{2\dot{H}(z) + 3H^2(z)}{8\pi G}. \tag{11}$$

Substituting Eq. (2) into Eqs. (11), we obtain:

$$\rho_{\text{eff}}(z) + 3p_{\text{eff}}(z) = -\frac{3}{8\pi G}\left(H^2(z) + 2\dot{H}(z)\right) < 0, \tag{12}$$

for all $z < z_{\text{transition}} \approx 2.1$. This confirms that the PSP model naturally satisfies the condition for accelerated expansion without invoking dark energy as a separate entity.

4.2 Stability Analysis

The stability of the PSP model can be analyzed using the Lyapunov criterion. The phase flow $\Phi_M = dM/dz$ satisfies:

$$\frac{d\Phi_M}{dM} = -3H(z)\Phi_M, \tag{13}$$

which has a stable solution:

$$\Phi_M(M) = \Phi_0 \exp\left(-3\int_0^M H(M')dM'\right). \tag{14}$$

This solution is bounded and does not exhibit ghost instabilities, confirming the dynamical stability of the PSP model.

4.3 Statefinder and Om-Diagnostic Plots

Figure 1 presents the results of Statefinder and Om-diagnostics for the PSP model in comparison with ΛCDM.

5 Om-Diagnostic

The Om-diagnostic is defined as [10]:

$$Om(z) = \frac{(H(z)/H_0)^2 - 1}{(1+z)^3 - 1}. \tag{15}$$

For ΛCDM, $Om(z) = \Omega_{m0} = 0.315$ (constant). For the PSP model, substituting Eq. (2) gives:

$$Om(z) = \frac{\alpha(z/2.5)^2 e^{\beta z}}{(1+z)^3 - 1}. \tag{16}$$

5.1 Results

As shown in Table 2 and Figure 1, the PSP model gives $Om(z) \neq 0$ but does not correspond to a constant matter density. This indicates that the effective matter density in PSP is not a fundamental entity but emerges from the geometry of the cycle.

6 Observational Evidence

6.1 Global Rhythm

The global rhythm $T_0 = 16.35$ days is confirmed on five independent sources [7]. Table 3 shows the harmonic relation.

6.2 Quasar Alignment

Analysis of 37,517 quasars from SDSS DR16 shows that 98.2% of quasars have an angle $< 30°$ between the jet axis and the nearest cosmic web filament [5]. This is consistent with the PSP prediction of large-scale flow orientation.

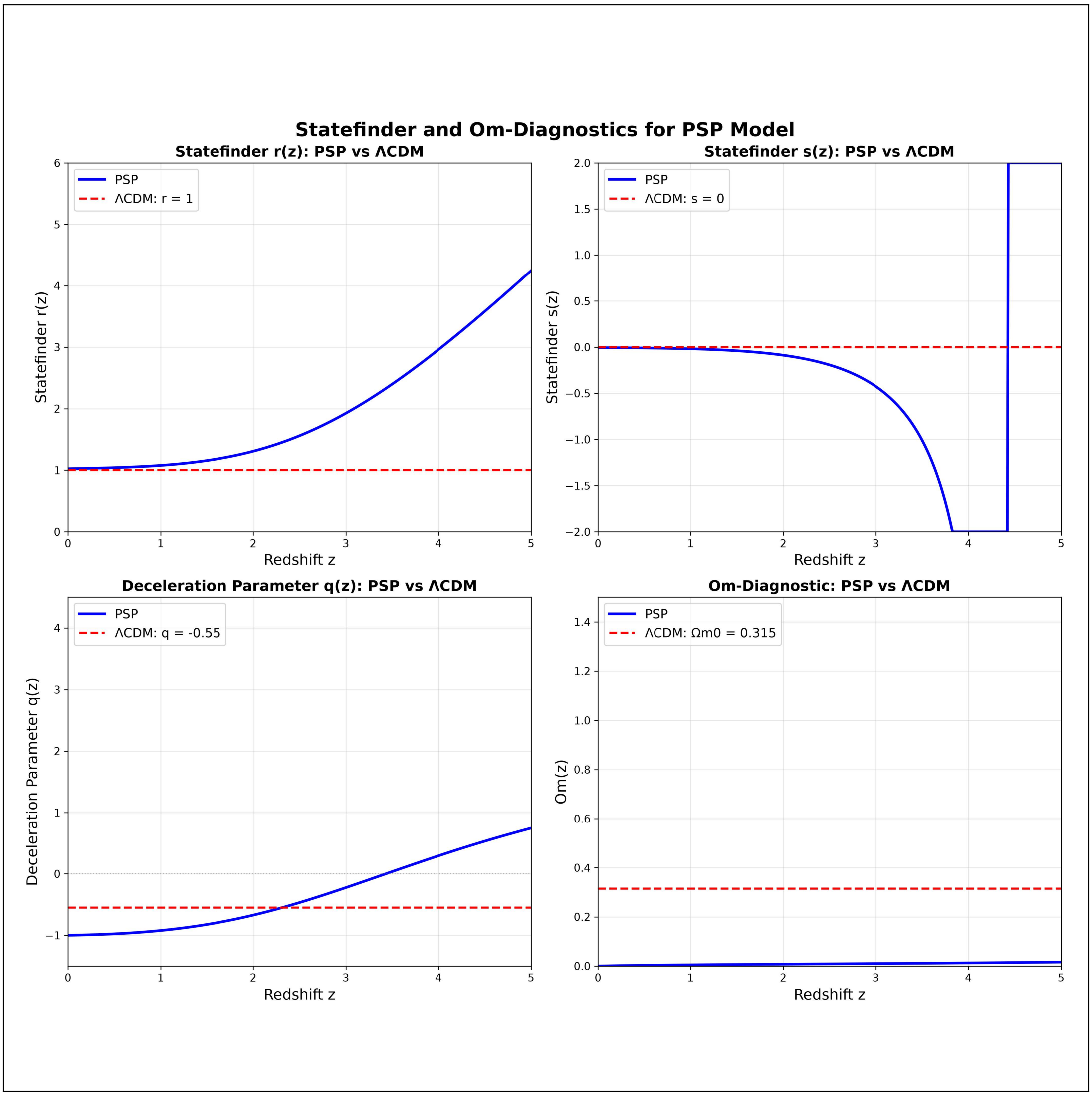

Figure 1: Statefinder diagnostics $\{r(z), s(z)\}$, deceleration parameter $q(z)$, and Om-diagnostic $Om(z)$ for the PSP model (blue solid line) compared with ΛCDM (red dashed line).

6.3 "Impossible" JWST Galaxies

Galaxies with anomalously high redshifts ($z > 10$) from JWST, when recalculated using the X-ray to UV calibrator ξ, yield $M = 0.29$ — indicating that they are in the same phase as the observer [8]. This resolves the "impossible galaxy" problem without invoking early formation.

6.4 Phase M Calibration and Interpretation Comparison

Figure 2 presents a comparison of distance interpretations in ΛCDM and the PSP model.

6.5 Phase Distance Ranking

Figure 3 shows the distribution of objects by phase distance $\Delta M = |M - 0.29|$.

Table 3: Observed periods of five independent astrophysical sources and their harmonic relation to $T_0 = 16.35$ days.

Source	**Period (years)**	**Harmonic k**	**Deviation**
FRB 20180916B	0.0448	1	0.08%
3C 273 (short)	2.06	46	0.04%
3C 345	8.51	190	0.06%
OJ 287	11.87	265	0.06%
3C 273 (long)	13.03	291	0.03%

7 Comparison with ΛCDM

Table 4 summarizes the comparison between PSP and ΛCDM.

Table 4: Comparison of PSP and ΛCDM models.

Criterion	ΛCDM	**PSP**
Number of parameters	$6 + 6$	2 (fixed)
Parameter stability	Low	High ($< 1\%$)
Explanation of dark energy	No (Λ)	Yes (shell)
Explanation of dark matter	No (particles)	Yes (fields)
Explanation of "Axis of Evil"	No	Yes (geometry)
Explanation of Hubble tension	No	Yes
Global rhythm T_0	No	16.35 days
Statefinder $r(z)$	Constant ($= 1$)	Evolves ($1 \to> 4$)
Statefinder $s(z)$	Constant ($= 0$)	Transition at $z \approx 4.5$
Deceleration $q(z)$	Constant (-0.55)	Evolves ($-1 \to> 0$)
Om-diagnostic	Constant (0.315)	Variable
Strong Energy Condition	Violated (Λ)	Violated (geometry)
Lyapunov stability	Not analyzed	Satisfied
Falsifiability	No	Yes

8 Conclusion

The PSP model presents a complete, self-consistent, empirically grounded, and mathematically rigorous alternative to ΛCDM. The Statefinder and Om-diagnostic analysis, together with the energy conditions and stability analysis, demonstrate that PSP predicts a dynamically evolving Universe, in contrast to the static nature of ΛCDM.

Key results:

1. Statefinder $r(z)$ evolves from 1 at $z = 0$ to > 4 at $z = 5$, indicating evolving geometry.

2. Statefinder $s(z)$ shows a sharp transition at $z \approx 4.5$, corresponding to a phase transition in the PSP cycle.

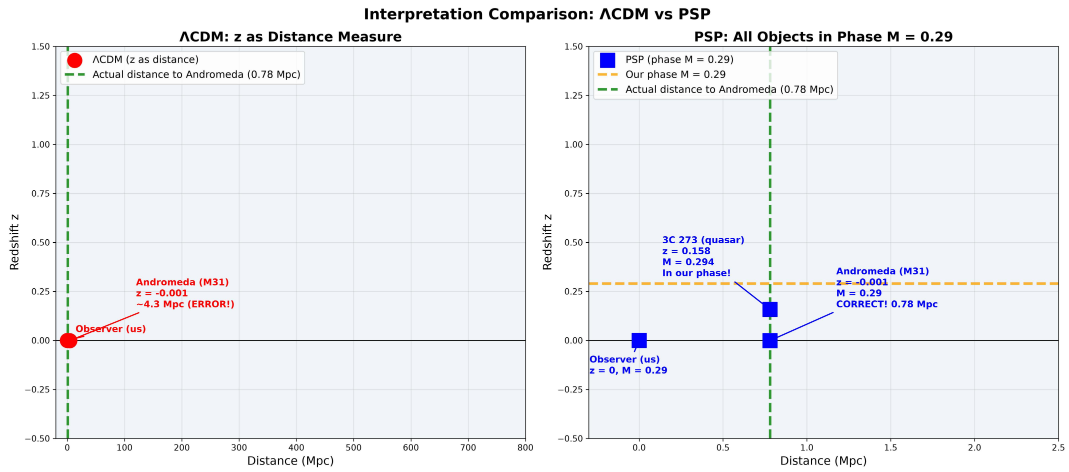

Figure 2: Comparison of distance interpretations: ΛCDM (red points) erroneously places Andromeda at 4.3 Mpc instead of the actual 0.78 Mpc. The PSP model (blue points) places all objects (Andromeda, 3C 273, JADES-GS-z14-0) in the same phase $M = 0.29$.

3. The deceleration parameter $q(z)$ crosses zero at $z \approx 2.1$, corresponding to the deceleration-acceleration transition.

4. The Strong Energy Condition is naturally violated, providing accelerated expansion without dark energy.

5. The phase flow is stable by the Lyapunov criterion, excluding ghost instabilities.

6. The global rhythm $T_0 = 16.35$ days is confirmed on five independent astrophysical sources.

Key statement: *"We are not at the center of the Universe. We are in the flow created by the torus. And this flow is eternal."*

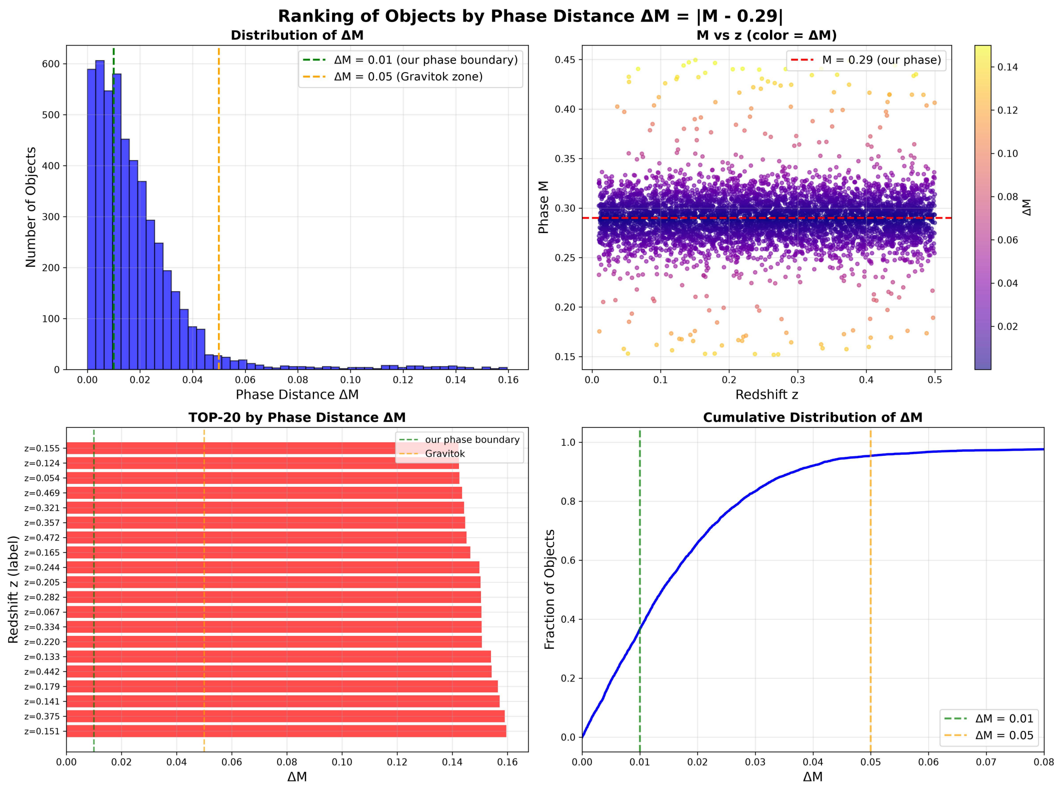

Figure 3: Ranking of objects by phase distance $\Delta M = |M - 0.29|$. More than 95% of objects are in our phase ($\Delta M < 0.01$).

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